

The Stranger Phone

Build guide and bill of materials — a vintage phone in a kaali-peeli that plays and records messages between strangers

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1. Concept recap

A vintage rotary or push-button telephone is mounted inside a Bombay kaali-peeli cab. A passenger notices a phone that clearly should not be there, picks up the handset, and an automated voice says "A stranger left a message for you", then plays a message recorded by a previous passenger. The voice then says "To leave a message for a stranger, press 1." Pressing 1 records the passenger's own message, which is added to the pool for future strangers.

The build rests on three pillars from the fieldwork: empathy (hearing a real stranger's voice), connection (a moment of contact in an otherwise isolated commute), and time (turning passive, zombie-like transit time into something meaningful).

2. How it works

Keep the vintage phone's shell, handset, cord and hook switch. Remove its original internals and put a small computer inside that does four things: notices when the handset is lifted, plays audio into the earpiece, records audio from the mouthpiece, and notices a "1" press. Everything else is software.

Hardware overview

The phone. A rotary or push-button Bakelite-style phone from Chor Bazaar or OLX. Push-button makes "press 1" trivial; rotary is more charming and still workable — dialling 1 sends exactly one pulse, which is counted on a GPIO pin. Only the shell, handset, cord and hook switch matter.

The brain. A Raspberry Pi Zero 2 W runs Python, plays and records audio, and sips power. A Pi 4 gives headroom for on-device speech-to-text or a web dashboard but roughly triples power draw.

Audio in/out. The Pi has no mic input, so a USB sound card (simplest) or an I2S audio HAT is added. Old handsets use a carbon microphone that will not work with modern inputs, so the mouthpiece capsule is swapped for an electret capsule. The earpiece speaker usually works as-is and is driven through a tiny PAM8403 amplifier so it is loud enough over cab noise.

Switches. The hook switch wires to one GPIO pin with a pull-up resistor. On a push-button phone the keypad matrix is scanned from a few GPIO pins, or the "1" key alone is wired as a single button. On a rotary phone one GPIO counts pulses.

Power. Two options, detailed in Section 4: a power bank (no wiring into the cab, no permissions, the driver swaps or charges it), or the cab's cigarette-lighter socket through a UPS HAT so the Pi survives ignition on/off cycles. In both cases the Pi's filesystem is set to read-only overlay mode with a separate writable partition for messages, so hard power cuts do not corrupt the SD card.

Mounting. Vibration and theft are the real enemies. Bolt the phone base to the dashboard or centre-console cover, or to a small plywood plate cable-tied to the seat frame, and keep the Pi in a case inside the phone body. The driver is the caretaker.

Software

Raspberry Pi OS Lite and a single Python script that is essentially a state machine:

- Idle → handset lifted (GPIO) → play greeting ("A stranger left a message for you")
- Pick one message from the local folder (random, or oldest-unheard) and play it
- Play prompt ("To leave a message for a stranger, press 1") and wait up to ~8 seconds for a 1
- Beep, record until hang-up, silence, or 60 seconds
- Play "Your message will find a stranger" and return to idle on hang-up
- Hanging up at any point stops everything and resets

Libraries: gpiozero for the hook and button, sounddevice or plain arecord/aplay for audio, pydub for trimming silence and normalising volume so quiet whispers and loud passengers sound similar. Prompts should be pre-recorded WAV files, not live text-to-speech — record them with a real human voice in Hindi, Marathi and English (or generate once and save), so nothing needs internet in the cab. Seed the phone with 15–20 messages recorded by the team and friends before it goes into a cab, so the very first passenger has something to hear.

Two decisions to make early

Moderation. Messages recorded by strangers will be played to strangers, so a buffer is needed. Simplest: new recordings land in a "pending" folder and only move to "approved" after someone on the team listens. With a phone hotspot or 4G dongle the Pi can upload recordings to a small server (Supabase, Firebase or a plain VPS) for browser-based approval; this also lets multiple cabs share one pool of messages and provides data for the showcase. Whisper transcription plus a text filter on the server can auto-flag the obvious cases.

Noise. Kaali-peeli cabs are loud. The handset's shape helps because the mouthpiece sits close to the mouth; add a noise gate in software and cap message length. Test in a moving cab early, not on a desk.

Fallback build

If the Pi route stalls: hide an old Android phone inside the vintage shell and wire the handset as a wired headset through a TRRS jack — earpiece and electret mic on the audio lines, hook switch wired like the headset play/pause button. A kiosk-mode web app does the rest, and battery, charging, 4G and a mic amp come for free. Less "hardware", but a two-evening build and a solid plan B before Goa.

Showcase tie-in

The Pi can log every play and record event, so a TouchDesigner visual at the showcase — a slowly growing map of message-to-stranger connections, or a live feed of message waveforms — lets visitors see the connection pillar, not just hear it.

3. Bill of materials — shared parts (both options)

Prices are approximate Indian street prices (Robu.in, Amazon.in, Lamington Road) as of September 2026 — check current listings before ordering.

#	Item	Qty	Approx. ₹	Notes / source
1	Vintage rotary or push-button phone (working handset, cord, hook switch)	1	1,500–4,000	Chor Bazaar / OLX. Push-button is easier for "press 1"
2	Raspberry Pi Zero 2 W (with header pins soldered)	1	2,400–2,800	Robu.in / Amazon.in
3	microSD card, 32 GB, A1-class	1	400–600	SanDisk / Samsung
4	USB sound card (mic in + headphone out)	1	300–800	Any CM108-based dongle works
5	Micro-USB OTG adapter (Pi Zero → USB-A)	1	100–150	For the sound card
6	Electret mic capsule, 6 mm, with leads	2	50 each	One spare. Replaces the carbon mic in the mouthpiece
7	PAM8403 mini amplifier module	1	80–120	Drives the earpiece from the sound card headphone out
8	10 kΩ resistors (pull-ups), 100 nF capacitors (debounce)	few	50	Lamington Road assorted pack
9	Tactile push button 12 mm (only if wiring a single "1" button instead of the keypad)	2	30	Optional
10	3.5 mm TRRS breakout or two 3.5 mm plugs	2	80	Connects handset wiring to the sound card
11	Dupont jumper wires + hookup wire (22 AWG)	1 set	150	
12	Heat-shrink tubing, small assorted	1	80	
13	Double-sided foam tape / hot glue sticks	1	100	Mounting the Pi and boards inside the shell
14	Small plywood/acrylic base plate + M4 bolts, cable ties	1	300	Mounting the phone in the cab
15	Micro-USB cable, right-angle, 1 m	1	150	Pi power lead
	Shared subtotal		≈ 6,000–9,500	

Tools needed but not per phone: soldering iron and solder (≈₹600), multimeter (≈₹500), wire strippers, a small screwdriver set.

4. Power options

Option A — Power bank

#	Item	Qty	Approx. ₹	Notes / source
A1	Power bank, 20,000 mAh, with low-current / "always-on" mode	1	1,500–2,500	Critical: many banks auto-shut-off below ~100 mA, and an idle Pi Zero sits at that threshold. Look for "trickle charge", "low-current mode" or "always on" (Anker, some Ambrane / Stuffcool models)

#	Item	Qty	Approx. ₹	Notes / source
A2	Power bank, 10,000 mAh, same requirement (spare / swap unit)	1	800–1,200	Lets the driver swap rather than wait for a charge
A3	Short USB-A → micro-USB cable, 30 cm	1	100	Keeps the bank inside or under the phone
A4	Velcro strap / small pouch to hold the bank	1	100	
	Option A subtotal		≈ 2,500–3,900	
	Option A total per phone		≈ 8,500–13,400	Shared parts + Option A

Expected runtime: the Pi Zero 2 W plus sound card and amp averages roughly 1–1.5 W, so a 20,000 mAh bank (~60 Wh usable) gives about 40–55 hours — two working days between charges. Set the Pi to a read-only overlay filesystem so a bank running flat does not corrupt the card. No wiring into the cab means no permissions and no driver hesitation.

Option B — Cigarette-lighter supply + UPS

#	Item	Qty	Approx. ₹	Notes / source
B1	Car cigarette-lighter USB adapter, 5 V ≥ 2.4 A (dual port is handy)	1	300–700	Branded (Portronics, Anker, Baseus); no-name ones put noise on the audio line
B2	Waveshare UPS HAT (C) for Pi Zero	1	2,000–2,600	Sits under the Pi Zero, USB-C input, powers the Pi and charges the cell simultaneously; reports battery level over I2C
B3	18650 Li-ion cell, 2,600–3,400 mAh, protected (Samsung / LG / Molicel)	1	300–500	Goes into the UPS HAT holder. Buy from a reputable seller
B4	USB-A → USB-C cable, 1.5–2 m	1	200–300	Lighter socket to UPS HAT; route along the dash edge
B5	Female / stacking header for Pi Zero	1	60	Only if the Pi Zero came without one
B6	Cable clips / spiral wrap for the cable run	1	100	
B7	Inline USB fuse	—	—	Not needed — the lighter adapter is already fused
	Option B subtotal		≈ 3,000–4,300	
	Option B total per phone		≈ 9,000–13,800	Shared parts + Option B

The 18650 cell in the UPS HAT gives roughly 8–10 hours of standalone runtime, so the phone keeps working when the driver stops for chai or the engine is off, and the Pi never sees a hard power cut when the ignition cycles. The UPS also allows a scripted clean shutdown when the cell drops below ~10 %. A plain 12 V→5 V buck module plus a pass-through power bank is a cheaper alternative, but it brings back the auto-shut-off problem from Option A, so the HAT is the cleaner route.

Practical note: some older kaali-peelis have a dead or unreliable lighter socket. Check the specific cab before committing — if it is dead, Option A is the fallback, and since the two builds differ only in the power section, nothing else changes.